

# DNS & Route 53

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## 1. DNS Basics

### 1.1 What is DNS and why is it important?

DNS (Domain Name System) resolves human-readable domain names to IP addresses.

Example: When you type `www.google.com`, DNS converts it to an IP like `142.250.190.78`, allowing your browser to reach Google.

### 1.2 Types of DNS Records

- A Record: Maps domain to IPv4. Example: `example.com` → `192.168.1.1`
- AAAA Record: Maps domain to IPv6. Example: `example.com` → `2606:4700:4700::1111`
- CNAME: Alias for another domain. Example: `blog.example.com` → `example.com`
- MX Record: Specifies mail servers. Example: `example.com` → `mail.example.com` (Priority: 10)
- TXT Record: Stores text data. Example: `v=spf1 include:_spf.google.com ~all`

### 1.3 Recursive vs. Iterative DNS Queries

Recursive Query: DNS server fully resolves the request. Example: Your computer asks a resolver for `example.com`, and it gets the final IP.

Iterative Query: DNS server returns a referral. Example: A DNS server may respond, 'I don't know, but try this other server.'

### 1.4 Primary, Secondary & Stub Zones

Primary Zone: Editable, stores DNS records.

Secondary Zone: Read-only copy of Primary.

Stub Zone: Stores only Name Server (NS) records.

### 1.5 Forward & Reverse DNS Lookup

Forward Lookup: Domain → IP. Example: `example.com` → `192.168.1.1`

Reverse Lookup: IP → Domain. Example: `192.168.1.1` → `example.com`, uses a PTR record.

## 2. DNS Server & Configuration

### 2.1 What is a DNS Resolver?

A DNS Resolver is the first DNS server that handles queries. Example: Your ISP's resolver translates `example.com` into an IP address.

## 2.2 What is TTL (Time to Live) in DNS?

TTL determines how long a record is cached before refreshing. Example: If TTL = 3600 seconds (1 hour), DNS caches the record for 1 hour before checking for updates.

## 2.3 What is Split-Horizon DNS?

Different DNS responses for internal and external users.

Example:

- Inside the company: intranet.example.com → 10.0.0.1
- Outside the company: intranet.example.com → No record found.

## 2.4 How do you flush the DNS cache on Windows?

Run the command:

'''

```
ipconfig /flushdns
```

'''

## 2.5 How do you test DNS resolution?

Use:

'''

```
nslookup example.com
```

```
Resolve-DnsName example.com
```

'''

# 3. DNS Troubleshooting & Security

## 3.1 How to troubleshoot DNS issues?

- Check network: ping google.com
- Test DNS resolution: nslookup example.com
- Verify settings: ipconfig /all
- Restart DNS service: Get-Service DNS | Restart-Service
- Flush DNS cache: ipconfig /flushdns

## 3.2 What is DNS Poisoning & Prevention?

DNS Poisoning injects fake records into a DNS cache, misdirecting traffic.

Example: A hacker makes facebook.com resolve to a malicious IP.

### **Prevention:**

- Use DNSSEC for cryptographic validation.
- Use DoH/DoT (DNS over HTTPS/TLS) for encryption.
- Reduce DNS cache TTL.

### 3.3 What is DNSSEC?

DNSSEC (DNS Security Extensions) adds cryptographic signatures to DNS records to prevent tampering.

Example: Verifying that example.com really belongs to the correct owner.

### 3.4 How to change the DNS server on Windows?

Run:

...

```
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses  
("8.8.8.8","8.8.4.4")
```

...

This changes your DNS to Google DNS (8.8.8.8, 8.8.4.4).

## DNS & Route 53 with Load Balancer

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### 1. DNS Setup with Load Balancer (Step-by-Step)

explains how to register a domain, configure Route 53, and point it to a Load Balancer for hosting a scalable and fault-tolerant website.

#### Step 1: Register a Domain Name

Use a domain registrar such as GoDaddy, Namecheap, or AWS Route 53.

Example: Register 'myappwebsite.com' via AWS Route 53 domain registration service.

#### Step 2: Deploy Your Application Behind a Load Balancer

Instead of pointing to a single EC2 instance, use an AWS Elastic Load Balancer (ELB) to distribute traffic across multiple instances.

Example: Deploy your web app in a target group behind an Application Load Balancer (ALB).

#### Step 3: Create a Hosted Zone in Route 53

In AWS Route 53, create a public hosted zone for 'myappwebsite.com'. This will allow you to manage DNS records.

#### Step 4: Update Domain's Name Servers

Copy the NS records from the hosted zone.

Go to your domain registrar (e.g., GoDaddy) and replace the default name servers with AWS Route 53 NS records.

#### Step 5: Create DNS Records Pointing to Load Balancer

- Create an 'A' record (Alias) in Route 53 hosted zone.
  - Choose Alias → Yes, and point it to the ALB DNS name (e.g., my-load-balancer-1234567890.us-east-1.elb.amazonaws.com).
- This allows your domain (myappwebsite.com) to route traffic to the load balancer.

#### Step 6: Add CNAME or Subdomain Records (Optional)

- Create a CNAME record for www.myappwebsite.com pointing to myappwebsite.com or directly to the ALB.
- Useful for supporting both www and root domain.

#### Step 7: Test Your Setup

Open a browser and type 'myappwebsite.com'. The request should reach the AWS ALB and route to healthy targets (EC2 or containers).

You can also test using 'nslookup myappwebsite.com' or 'dig myappwebsite.com'.

#### Optional Step 8: Use Cloudflare or Azure DNS

- Add your domain to Cloudflare.
- Replace NS records at the registrar to use Cloudflare's DNS.
- Create a CNAME pointing to the AWS ALB DNS and enable SSL, caching, and WAF features.